



EMPIRICAL RESEARCH

Mindful revolution or mindless trend? Examining agile development as a management fashion

W. Alec Cram¹ and
Sue Newell²

¹Information and Process Management, Bentley University, Waltham, U.S.A.; ²University of Sussex, Brighton, U.K.

Correspondence: W. Alec Cram, Information and Process Management, Bentley University, 175 Forest Street, Waltham MA 02452 U.S.A.
E-mail: wcram@bentley.edu

Abstract

The growing adoption of agile methods of system development can be seen as broadly consistent with characteristics of a management fashion. However, the extent that fashions relate to how specific organizations actually adopt and use agile remains unclear. Based on five case studies, this research seeks to apply the concepts from management fashion theory to understand how organizations mindfully (or mindlessly) adopt agile methods. We find in some cases that agile appears to be characterized as a short-term, transitory trend, while in others it is embedded in an organization's processes and culture. We identify three categories that distinguish between fashion patterns during agile adoptions: Crusaders, who exclusively adopt agile in a pure form; Tailors, who integrate agile and traditional approaches to fit their specific circumstances; and Dabblers who employ a few ceremonial agile activities alongside a traditional approach. The results from this study can encourage managerial awareness of the link between fashions and agile development, which can serve to highlight the unique opportunities to enhance mindfulness and avoid the pitfalls of mindlessness. By uncovering insights into the fashion-centric drivers of agile adoption, this research extends past theory related to both mindful innovation and management fashions.

European Journal of Information Systems (2016) 25(2), 154–169.

doi:10.1057/ejis.2015.13; published online 19 January 2016

Keywords: agile; systems development; management fashion; mindfulness; adoption

Introduction

The increasingly mainstream adoption of agile software development in recent years has resulted in many organizations transitioning away from a plan-driven, waterfall approach (Cao *et al.*, 2009; De Cesare *et al.*, 2010; West & Grant, 2010). Agile refers to a family of iterative systems development methods that value team collaboration, minimal up-front planning, and the flexibility to adapt to changing requirements (Beck *et al.*, 2001). Each organization's decision to adopt agile stems from a variety of factors, including the desire to improve efficiency, negative experiences with other development approaches, and pressure from stakeholders to adopt innovative development approaches (Mangalaraj *et al.*, 2009; Maruping *et al.*, 2009; Senapathi & Srinivasan, 2012).

Although the perceived benefits of agile have been widely reported (Chow & Cao, 2007; De Cesare *et al.*, 2010), opposing views that challenge the hype surrounding agile have noted examples of deployment difficulties, ambiguous benefits, and lack of project ownership (Balijepally *et al.*, 2009; Laanti *et al.*, 2011;

Received: 4 October 2014
Revised: 3 March 2015
2nd Revision: 20 July 2015
3rd Revision: 13 August 2015
Accepted: 18 August 2015

Drury *et al.*, 2012; Janes & Succi, 2012). In addition, many organizations are avoiding a 'by the book' adoption of agile and are instead customizing agile techniques (Fitzgerald *et al.*, 2006; Conboy & Fitzgerald, 2010) or employing agile alongside other development methods (Cao *et al.*, 2009; Port & Bui, 2009; West & Grant, 2010). Although past research has investigated the practicalities of agile customization and adaptation, it remains unclear what factors are driving this adoption variability and how the use of agile techniques differs from project to project.

This research positions management fashion theory as one possible explanation by suggesting that management techniques created by consultants, authors, and academics are continually disseminated to organizations and are presented as novel approaches to drive performance and innovation; however, after a period of rising popularity, the techniques are eventually abandoned in favor of a newly emerging fashion (Abrahamson, 1991; Abrahamson, 1996; Abrahamson & Fairchild, 1999). Past research has applied a fashion theory lens to general management techniques such as quality circles and total quality management, as well as information systems techniques such as service-oriented architecture and computer-aided software engineering (Lawler & Mohrman, 1985; Baskerville & Myers, 2009; Hirschheim *et al.*, 2012). We suggest that at least some aspects of management fashions are present in virtually all adoptions of a new systems development approach, such as the belief that an approach is highly efficient or advanced (Abrahamson, 1996; Abrahamson & Fairchild, 1999). However, rather than simply evaluating if agile is a management fashion, this research seeks to use the concepts from management fashion theory to understand how organizations adopt and use agile methods.

A key factor driving the acceptance of an innovation is the degree that its adoption exhibits characteristics of mindfulness (e.g. reluctance to simplify complex situations, vigilance in attending to small operational details) or mindlessness (e.g. lack of attention to specifics, contentedness to be a follower) (Fiol & O'Connor, 2003; Swanson & Ramiller, 2004). Mindful approaches to innovation represent an opportunity to drive improved performance, support improved decision making, and improve the speed of issue detection (Butler & Gray, 2006). Although the development practices that are consistent with a mindful agile adoption have been examined in past research (Matook & Kautz, 2008; Vidgen & Wang, 2009; McAvoy *et al.*, 2013), it remains unclear what factors actually influence a company to mindfully (or mindlessly) adopt agile, as well as the practical impact that these different approaches have on the ongoing use of the innovation. As a first step in addressing this gap, this research considers the role of management fashions.

We pose two related research questions: *How do macro-level management fashion indicators relate to firm-level adoptions of agile development? Do variations in the impact of different management fashion indicators at the firm level relate to whether agile adoptions can be characterized as mindful or mindless?* Using 42 interviews with systems developers and

managers at five organizations, we employ a qualitative case study approach to analyze the perspectives of those involved in the adoption and use of agile. Drawing on past research, we investigate five factors that indicate the existence of a management fashion: norms of progress, norms of rationality, socio-economic forces, techno-economic forces, and innovation devolution (Abrahamson, 1996; Zbaracki, 1998; Abrahamson & Fairchild, 1999; Carson *et al.*, 1999; Benders & van Veen, 2001). Based on patterns within the fashion indicators across the cases, we assess the extent that an innovation adoption can be characterized as mindful or mindless in order to examine the opportunities and risks related to agile adoptions.

The adoption and use of agile has been steadily increasing over the past decade, positioning it as a fundamentally important driver of productivity and performance within many of today's IT departments (West & Grant, 2010). Even if this momentum wanes in the future, organizations remain concerned with making systems development choices today that enhance their future opportunities for projects that are on time, in scope, and within budget. One key decision, the adoption and use of a particular development approach, is commonly seen to be driven by project-level factors such as team size or system complexity (Boehm & Turner, 2003; Nerur *et al.*, 2005; Vinekar *et al.*, 2006). However, additional macro-level factors such as management fashions may also influence these decisions. Such technical innovations can aid organizations in improving their reputation and long-term performance (Wang, 2010), but may also encourage inefficiency, deplete competitiveness and restrict organizational learning (Abrahamson, 1991; Hirschheim *et al.*, 2012). By considering how fashions relate to the adoption and use of agile, as well as the corresponding degree of mindfulness, this research can help organizations reflect on what they seek to gain from agile methods and how institutionalized agile should become in their processes. These longer-term perspectives can aid managers in extending beyond a short-term, project-by-project view of agile to increasingly consider the strategies and trends that can be leveraged in relation to agile development.

The remainder of the paper is as follows. In the following section, we provide an overview of agile development and management fashion theory. Next, we outline our methodology, including the data collection and analysis approach. The results are then presented and discussed, including the connection with mindful innovations. We conclude by presenting a series of future research opportunities.

Background and theoretical base

This research focuses on agile systems development approaches in the context of management fashions. We first outline the core concepts and history of agile development and then highlight the fundamentals of management fashion theory. These two elements are integrated through the argument that management fashions can help us better understand organizational adoption and use of

agile. Based on this foundation, we then present a framework of five management fashion indicators that form a base for our analysis of the empirical data.

Agile development

Agile development refers to a collection of software development practices that fundamentally differ from a plan-driven, waterfall approach. Widespread awareness of agile originated in 2001 with the release of the Agile Manifesto by a group of 17 practitioners who sought to establish a better way of developing software. The Manifesto established a core philosophy based on the importance of individuals and interactions, the incremental delivery of working software, collaboration with customers, and responding to changes (Beck *et al.*, 2001). Twelve underlying principles were included in the manifesto, including the use of face-to-face conversations, maintaining a constant pace, technical excellence, and self-organizing teams.

A variety of specific methods that operationalize and provide guidance on the agile philosophy include eXtreme Programming (XP), Scrum, Crystal Methods, and Dynamic Software Development Method (DSDM). These methods prescribe specific practices, such as pair programming, daily stand-up meetings, shared code ownership, and retrospective project evaluations. Past research in the field has examined the effectiveness and benefits, as well as the practical challenges, of adopting such practices (Balijepally *et al.*, 2009; Maruping *et al.*, 2009; Drury *et al.*, 2012).

In considering the rise of agile development as an alternative to more traditional development methods, a key sub-category of agile development research examines the adaptation of the approach (Cao *et al.*, 2009; Port & Bui, 2009; Conboy & Fitzgerald, 2010). Organizations can tailor their development methodologies by adopting a contingency approach (e.g. one of a collection of possible methods is chosen, depending on the project context), a method engineering approach (e.g. methods are constructed from a set of pre-defined method components), or an *a la carte* approach (e.g. fragmented elements of a method or group of methods are selected for use together) (Fitzgerald *et al.*, 2006). This line of inquiry is of particular interest to those organizations that perceive value in the agile approach, but are reluctant to fully abandon traditional waterfall techniques or follow a prescribed set of guidelines. Research suggests that the initial adoption of agile is driven by considerations including project size, application criticality, complexity, employee skillset, and company culture (Boehm & Turner, 2003; Nerur *et al.*, 2005; Vinekar *et al.*, 2006). In situations where agile is employed alongside a traditional development approach, challenges related to conflicting values, unclear employee expectations, and confused management styles can surface (Nerur *et al.*, 2005; Vinekar *et al.*, 2006). Although evidence exists that some organizations may be able to adopt agile techniques alongside techniques from other methodologies (Boehm & Turner, 2003; Holmstrom *et al.*, 2006; Port & Bui, 2009; Vinekar *et al.*, 2006), other research

suggests that agile techniques that are adapted to better suit a development process, developer style, customer, and organization may ease an organization's transition away from a traditional approach (Fitzgerald *et al.*, 2006; Cao *et al.*, 2009). Characteristics such as a description and rationale for method practices and a familiarity with a range of methods have been shown to improve the effectiveness of methodology tailoring (Conboy & Fitzgerald, 2010).

Despite the growing use of agile in practice, academic research was initially slow to respond and has only picked up in recent years (Dyba & Dingsoyr, 2008; Abrahamsson *et al.*, 2009). Special journal issues, such as the *European Journal of Information Systems* (2009) and *The Journal of Systems and Software* (2012), have raised the profile of agile-oriented research. Although some work in the field has been criticized for lacking a strong theoretical and conceptual base (Conboy, 2009; Wang *et al.*, 2012), the variety of theoretical perspectives is growing, with knowledge management and personality-related foundations being the most prevalent (Dingsoyr *et al.*, 2012). Some suggest that agile is simply a passing fashion (Sharp *et al.*, 2006; Janes & Succi, 2012; Moczar, 2013), while others suggest that it is here to stay because of its superior speed and flexibility relative to alternatives (Nerur *et al.*, 2005; Holmstrom *et al.*, 2006; De Cesare *et al.*, 2010; Dingsoyr *et al.*, 2012). However, rather than debating if agile is broadly characteristic of a fashion, we consider the possibility that this might be a firm-dependent variable, with some organizations adopting agile in a way that suggests the approach is merely a transitory phase, while in other organizations agile is adopted in a way that suggests it is becoming or already is a permanent (and institutionalized) approach. We consider this possibility through the lens of management fashion theory, which is discussed next.

Management fashion theory

This research considers the growth of agile development in the context of management fashion theory, which refers to transitory beliefs that particular management techniques lead to a rational management progress (Abrahamson, 1991; Abrahamson, 1996; Abrahamson & Fairchild, 1999). Consistent with Abrahamson & Fairchild (1999) and Baskerville & Myers (2009), this definition is neutral on the merit of management fashions and remains open to the possibility of positive and/or negative consequences from adopting a fashion. Building on the theory of adoption and diffusion (Rogers, 1995), management fashion theory argues that new organizational techniques are influenced by factors including the rhetoric of influential gurus, research funding bodies, and practitioner demand for innovative solutions to practical problems (Abrahamson, 1991; Abrahamson, 1996; Abrahamson & Fairchild, 1999; Newell *et al.*, 2001). Fashions are initiated by organizations and individuals (e.g. consultants, gurus, authors, academics) who continuously create and distribute management knowledge. As this knowledge is disseminated to organizations, some managers (though not

all) will perceive the new techniques as being superior to the current techniques and feel compelled to iteratively adopt a new approach (Abrahamson, 1991; Abrahamson, 1996), which is subsequently replaced as new fashions emerge.

Examinations of management fashions point to a process whereby a management technique is invented (or reinvented), a period of latency, a popularity surge, then a gradual falling out of favor (Kieser, 1997; Abrahamson & Fairchild, 1999; Carson *et al*, 1999). In contrast to popular perceptions of clothing fashions, which may only last a year or two, the lifecycle of a management fashion can extend for a period of 20 years (Abrahamson & Fairchild, 1999). Although this perspective can be helpful in examining macro-level trends, it is unclear if applying a management fashion lens can be helpful in better understanding what influences firm-level adoptions, such as agile development.

A variety of internal and external triggers can shape management fashions, including perceived consistency with organizational structures, the organizing vision of the innovation, individuals' psychological needs, or the emergence of a new fashion (Abrahamson, 1996; Swanson & Ramiller, 1997; Abrahamson & Fairchild, 1999). The rise of a new management fashion can be triggered in circumstances such as when an existing management fashion is no longer viewed as being effective or another management technique is perceived as needing to address a performance gap (Abrahamson & Fairchild, 1999). If a fashion rises in popularity but the anticipated performance improvements are not achieved, counter-fashions may emerge that reverse the approach of the initial technique (e.g. decentralization to centralization, outsourcing to insourcing) (Kieser, 1997).

Past investigations of general management fashions consider a range of trends including business process re-engineering, total quality management, and quality circles (Newell *et al*, 1998; Zbaracki, 1998; Abrahamson & Fairchild, 1999). Although these studies recognize the negative implications of adopting management fashions, such as restricted organizational learning, lower short-term performance, and inefficiency, positive outcomes are also noted including improved organizational reputation, long-term performance, and facilitating political manoeuvres (Abrahamson, 1991; Kieser, 1997; Newell *et al*, 1998; Baskerville & Myers, 2009; Wang, 2010; Hirschheim *et al*, 2012). In fact, some research suggests that transitory management fashions are more pervasive within organizations than stable, institutionalized techniques (Abrahamson & Fairchild, 1999). As a result, we avoid characterizing an organization's adoption of a management fashion as either 'good' or 'bad'; evidence exists that fashions can contribute to both positive and negative organizational outcomes. Rather, we seek to consider how the elements that indicate the existence of a management fashion (discussed below) relate to firm-level adoptions, in order to evaluate if an adoption can be characterized as mindful or mindless and pervasive or transitory, recognizing that perhaps these aspects of adoption are not necessarily tightly related.

The concept of mindfulness builds on social psychology principles to aid in better understanding how individuals process information and behave in social situations (Langer *et al*, 1978; Langer, 1989b). Where a state of mindfulness exists, qualities of openness and alertness are present, which is in contrast to a mindless state of 'tuning out' and ignoring new information (Langer *et al*, 1978; Langer, 1989a; Langer, 1997). When considered within an organizational and information systems setting, the role of mindfulness has been applied to topics including learning (Levinthal & Rerup, 2006; Weick & Sutcliffe, 2006), innovation (Swanson & Ramiller, 2004; Mu & Butler, 2009), and reliability (Vogus & Welbourne, 2003; Butler & Gray, 2006). In such contexts, mindful actions may take the form of a reluctance to simplify complex situations and a vigilance in attending to small operational details, while mindless actions demonstrate a lack of attention to specifics and contentedness to be a follower (Swanson & Ramiller, 2004).

Within the information systems discipline, topics such as e-commerce, enterprise resource planning systems, service-oriented architecture, knowledge management, group decision support systems, and computer-aided software engineering have been investigated in the context of management fashion theory (Westrup, 2003; Ramiller *et al*, 2008; Baskerville & Myers, 2009; Hirschheim *et al*, 2012). This body of research suggests that the fundamentals of management fashions apply within IS, though distinctions such as higher switching costs associated with investments in hardware and software make IS fashions somewhat unique (Baskerville & Myers, 2009; Wang, 2010). Building on Abrahamson's definition, an IS fashion is defined as 'a relatively transitory collective belief in IS research and practice, disseminated by fashion setters, that a technique or technology leads to rational IS innovation' (Baskerville & Myers, 2009, p. 649).

Much of the past management fashion literature adopts a macro-level perspective that considers the trajectory along which a new technique is adopted or abandoned using approaches such as citation analysis (Abrahamson & Fairchild, 1999; Baskerville & Myers, 2009; Hirschheim *et al*, 2012). However, the extent that fashions relate to firm-level adoption of management techniques remains relatively unclear, despite calls for alternative approaches that adopt such perspectives (Zbaracki, 1998; Clark, 2004). We seek to address this gap by examining the links between management fashion concepts and the adoption of agile development methods.

Considering agile as a fashion that can impact mindfulness

Although management fashion theory is only one of many perspectives that can generate new insights into agile adoption and use (Dingsoyr *et al*, 2012), we suggest that it can contribute unique insights. Drawing on management fashion concepts to understand how organizations adopt and use agile can add a new perspective to the current literature, which tends to focus on employing

development methods on the basis of project-specific factors such as goals, size, customer relations, problem-solving approach, and learning type (Boehm & Turner, 2003; Nerur *et al.*, 2005; Nerur & Balijepally, 2007). Although we agree that these factors are key in aligning a development approach with the task at hand, fashion theory presents a collection of unique, macro-level factors that also influence firm-level development decisions. Since fashion theory focuses on specific concepts (e.g. rationality, rhetoric, psychological needs), it cannot fully explain all aspects of how agile is adopted and used. However, we believe that fashion theory presents a new, longer-term perspective that managers can consider when making decisions about the relative benefits of agile (e.g. are there other development approaches that are more suitable for the organization?), the ongoing use of agile (e.g. will we continue to use agile in five years?), and the extent that agile techniques are institutionalized (e.g. is agile our standard, accepted approach to development?).

We suggest that at least some general aspects of agile development are consistent with past management fashion characterizations. For example, the Agile Manifesto could be viewed as a widely cited rhetorical discourse that justifies and promotes the use of the agile approach. The broad awareness of the Manifesto is consistent with the view that management fashions are supported via a dissemination of knowledge from fashion-setting sources (Abrahamson & Fairchild, 1999) – in this case, prominent practitioners. This has been further supplemented by increasing academic and practitioner interest, as evidenced by journal special issues and publications in a range of technology and management outlets. This growing body of agile literature is consistent with past patterns of management fashions, which typically lag behind in generating attention from the popular press (Abrahamson & Fairchild, 1999). Benders & van Veen (2001) also point to an appealing name or label as being indicative of a management fashion. Since the term ‘agile’ is associated with speed and responsiveness, the approach is appealing to organizations.

In connecting these agile characteristics with management fashion theory, we adopt an exploratory approach that specifies the constructs of interest, but does not make predictions based on theory (Dube & Pare, 2003). Specifically, we draw on past research to identify a widely used collection of generic management fashion indicators and apply them to an agile development context (see Table 1). We posit that nearly all innovations that are adopted by organizations will contain at least some evidence of the indicators listed below. However, it is unclear if patterns within these fashion indicators may characterize a mindful vs mindless agile adoption and whether this is directly related to a transitory vs more institutionalized adoption at the firm level.

In total, five indicators are identified within the literature. The first indicator is the norm of progress, which suggests that new management techniques will be adopted because they are seen as *novel* innovations (Abrahamson, 1996). Due to ongoing demands for

improved performance, this indicator suggests that managers will choose techniques that appear to represent continuing improvement. In some cases, an innovation may be adopted based on the norm of progress, alongside supporting evidence and careful consideration that the advancement is genuine; however, in more transitory, fashion-centric scenarios, an innovation may be adopted principally due to the fact that it is perceived as being cutting edge, without careful consideration of the evidence to support such claims at the firm level. In either case, an organization’s engagement with an innovative technique can signal to stakeholders that progress and advancement is being made (Wang, 2010).

The second indicator, the norm of rationality, suggests that new management techniques will be adopted when they are presented by others as representing the *most efficient* approach available (Abrahamson, 1996). Drawing from Meyer and Rowan’s (1977) view of neo-institutional theory, this indicator assumes that managers and employees will choose to act in a rational manner by following the conclusions of outside experts. As gurus and academics promote a new management technique as being superior to previous approaches, the norms of rationality suggest that the new technique will be adopted in order to act in the best interests of the organization. Whereas the norm of progress indicator suggests that a technique is desirable solely on the basis that it is new, the norm of rationality indicator argues that a technique is desirable because it performs better than the alternatives. Although the norms of rationality and progress are viewed as being conceptually distinct, we recognize that some companies will adopt a technique on the basis that it is both new *and* more efficient than the alternatives. However, where accepting the performance conclusions of fashion-setters at face value may indicate a management fashion, this should be distinguished from a scenario where an organization adopts a management technique on the basis of fashion-setter conclusions, but also conducts its own independent analysis. In such cases, an organization may well be mindfully adopting a technique on the basis of its merits, rather than simply following the path advocated by others.

The socio-psychological forces indicator refers to the increased demand of a management innovation based on the perception that it *fulfills individuals’ psychological needs*, such as individuality, novelty, and status (Abrahamson, 1996). This indicator argues that managers will gravitate towards new techniques in order to deviate from the status quo and be perceived as being fashionable and progressive. In doing so, this indicator also encompasses an individual’s desire to avoid the anxiety caused by organizational performance gaps by suppressing these psychological pressures through exaggerating the potential value of the innovation as a quasi-magical and all-powerful solution for the organization’s problems (Abrahamson & Fairchild, 1999). Although such scenarios would be consistent with a management fashion, there may be other situations where an innovation is adopted on the basis that it is a genuine complement to the social and psychological characteristics of employees. Although such scenarios would remain

Table 1 Management fashion indicators

Management fashion indicators	Indicator description	Indicator source
Norm of progress	New management techniques will be adopted because they are seen as novel innovations.	Abrahamson, 1996; Carson <i>et al</i> , 1999
Norm of rationality	New management techniques will be adopted when they are presented by others as representing the most efficient approach available.	Abrahamson, 1996
Socio-psychological forces	The demand for management innovations will increase when they are perceived as fulfilling individuals' psychological needs.	Abrahamson, 1996; Abrahamson & Fairchild, 1999; Carson <i>et al</i> , 1999
Techno-economic forces	The demand for management innovations will increase when they are perceived as being consistent with economic fluctuations, political perspectives, and organizational structure or culture.	Abrahamson, 1996; Carson <i>et al</i> , 1999
Innovation devolution	A management technique that begins as an authentic technical intervention, but devolves into ambiguous and dubious inventions that provide little value aside from perceived legitimacy.	Zbaracki, 1998; Benders & van Veen, 2001

consistent with the management fashion indicator, it would suggest the technique's adoption would be increasingly consistent with mindful decision-making.

The fourth indicator, techno-economic forces, refers to the increased demand of a management innovation that is perceived as being *consistent with economic fluctuations, political perspectives, and organizational structure or culture* (Abrahamson, 1996; Carson *et al*, 1999). Where such management technique characteristics are perceived as complementing the internal characteristics of the organization, they become more desirable and are pursued more strongly, which can indicate consistency with a management fashion. Although this could suggest a temporary fashion that is rapidly passed over for another trend that follows a subsequent techno-economic fluctuation, an organization may rationally adopt an innovation due to longer-term, pervasive trends that would be less characteristic of transitory fashions.

Finally, the innovation devolution indicator refers to management techniques that begin as authentic technical interventions, but devolve (after adoption by an organization) into *ambiguous and dubious inventions* that provide little value aside from perceived legitimacy (Zbaracki, 1998). Where organizations adopt an innovation ceremonially, but fail to integrate the core elements, the technique will lose power and become consistent with a management fashion. This indicator also draws on the concept of interpretive viability (Benders and van Veen, 2001), which suggests that where there is some degree of conceptual ambiguity in a management technique, organizations can opportunistically select only those elements that apply to them and reinterpret the fashion's core idea to align with their own needs. Although interpretive viability can be applied as a driver for increased diffusion of a fashion, it can also apply as a driver of innovation devolution. In some cases, interpretive viability may be a response to an innovation that requires customization to align with an organization's internal or external characteristics and needs; however, when the innovation is adapted

to such a point that it begins to lose the spirit of its initial conceptualization, it devolves into a management fashion.

We argue that at least some of the five management fashion indicators discussed above are inherent in the adoption of most management techniques. However, the extent that these macro-level factors relate to firm-level adoptions is unclear. In response, this study examines whether management fashion concepts can aid in understanding if agile adoptions can be characterized as mindful or mindless and transitory or pervasive. In the next section, we outline the methodology employed in this research to address this gap.

Methodology

A common approach adopted in past management fashion studies is to examine the number of instances that a particular technique appears within the practitioner and academic press in order to evaluate its rise and fall in overall popularity (Abrahamson & Fairchild, 1999; Baskerville & Myers, 2009; Hirschheim *et al*, 2012). Although valuable, commentators have highlighted the additional insights that can be generated by examining the events occurring within specific organizations in response to these fashion-setting drivers related to the decisions to adopt, use, or abandon new techniques (Zbaracki, 1998; Benders and van Veen 2001; Clark, 2004). In examining these micro-level scenarios, past research has also recognized the appropriateness of qualitative studies to examine specific cases of the creation and selection phases of the management fashion process (Abrahamson, 1996; Zbaracki, 1998).

Data collection

In order to cultivate this rich viewpoint of management fashion concepts within firm-level agile adoptions, our study employs a qualitative approach based on 42 interviews with participants involved in systems development projects that include agile techniques. Five case study organizations (names have been disguised at the request

Table 2 Interviewee and organizational overview

	<i>LargeMan</i>	<i>InfoOrg</i>	<i>HealthOrg</i>	<i>InsureCorp</i>	<i>CodeCorp</i>
<i>Industry</i>	Manufacturing	Media	Healthcare	Insurance	Information Technology
<i>Number of interviews</i>	12	8	4	7	11
<i>Example interviewee titles</i>	Development Manager, Project Manager, Developer	IT Project Manager, Developer	Director of IT, Systems Development Manager	VP of Systems Development, Business Analysis Manager, Developer	VP of IT, Development Manager, Developer
<i>Style of agile approach</i>	Scrum	Scrum	Customized Scrum	Not formally defined	XP
<i>Initial date of agile adoption</i>	2009	2004	2011	2005	2004

of the participating organizations) were selected based on a theoretical replication strategy, which sought to uncover variations in results due to distinctions in the underlying site components (Yin, 2009). Variance was established in the agile method used (e.g. XP, Scrum), the duration that agile had been in place for, and the company's industry. Interviewees included executives (e.g. VP of Systems Development, VP of IT, Director of IT), middle managers (e.g. Systems Development Manager, Business Analysis Manager, Manager – Business Systems), and staff (e.g. Developer). Interviewees were selected on the basis of participation on a project that employed at least some agile techniques. We began our interviews with a senior executive and requested interviews with supplementary project team members. The data were collected as part of a broader research program investigating control within systems development processes. Details of the participants and organizations are noted in Table 2, including the industry, number of interviews, agile approach details, and date of agile adoption.

LargeMan is a multi-national manufacturing organization based in the United States. They have historically relied on a traditional waterfall development approach, but began to pilot agile techniques in 2009. By 2011, some project teams within their development process were exclusively using agile techniques based on Scrum. Since then, additional teams have adopted Scrum, though other projects continue to rely on a traditional approach. Our data were collected in two stages: in 2009 when agile was initially being piloted and again in 2011 when it was more fully rolled out. Interviewees belonged to project teams using agile within the organization's Canadian subsidiary. The techniques used by the team included daily stand-up meetings, story cards, and 3-week delivery cycles. The introduction of agile was primarily a grass-roots initiative from a small group of developers and was initiated largely on the basis of a perceived ineffectiveness of the waterfall technique.

InfoOrg is a large, U.S.-based organization that provides media and information services to customers around the world. The organization has historically relied on a traditional waterfall development approach, but has transitioned a collection of core projects to an agile approach that follows Scrum. Adopted practices

include stand-up meetings, pair programming, and limited documentation. The initial push to adopt an agile approach was initiated by an IT executive and interest in the approach has been growing steadily within the company.

HealthOrg is a large teaching and research hospital based in Canada. Strategically, the organization follows a best-of-breed approach to software, allowing clinical departments to purchase or build whatever systems they feel contribute most effectively to patient care. Some in-house development and customization is conducted on these clinical systems, as well as work on back-office administrative systems and integrating the systems together. HealthOrg has traditionally adhered to a structured, waterfall development approach, but recently began to use agile. The adopted approach is based in Scrum, but includes extensive customizations.

InsureCorp is a medium-sized insurance and financial services company based in Canada. The organization employs a mix of legacy and modern systems, including both off-the-shelf and custom developed applications. InsureCorp has traditionally followed a structured waterfall approach. In 2005/2006, a consulting firm was hired to assist the company in adopting agile, but the project was widely viewed as a failure and the company returned exclusively to waterfall shortly afterwards. In 2011, InsureCorp determined that they would make another attempt, but the previous experience has led to an avoidance of the term 'agile', because of the negative connotation within the organization. As such, no formal alliance with a specific agile method (e.g. Scrum, XP) has been established, although elements of these approaches are in place (e.g. sprints, stand-up meetings). Overall, InsureCorp is undertaking a highly cautious agile adoption. They view the approach as being a better way of expediting development, but are still learning about agile and are selecting only the techniques they perceive to be most valuable.

CodeCorp is a medium-sized IT services organization that specializes in developing custom software systems, primarily for financial services and healthcare clients. They are based in Canada and their systems development teams operate from a single office location. The organization has a long history with agile development and adopts an eXtreme

Table 3 Management fashion indicators for agile

Management fashion indicator	Indicator description
Norm of progress	Agile is adopted by organizations because it is perceived as a novel technique. For example, agile is adopted because it is viewed as the most cutting-edge, advanced approach to systems development.
Norm of rationality	Agile is adopted based on pronouncements by others that it provides the most efficient means to develop systems. For example, the adoption of agile is influenced by a development manager who reads an article by a prominent author that argues agile is a faster development approach compared to a plan-based waterfall approach.
Socio-psychological forces	Agile is perceived to fulfill the psychological needs of employees (e.g. individuality, novelty, avoidance of anxiety). For example, agile is adopted because it is viewed by developers as enabling them to be more creative and feel that they are contributing to the project deliverables.
Techno-economic forces	Agile is perceived to be consistent with industry and organizational characteristics such as economic fluctuations, politics, structure, and culture. For example, agile is adopted because it is viewed as being an appropriate fit for a company's collaborative culture.
Innovation devolution	Agile is sufficiently ambiguous to allow it to be ceremonially adopted in a manner that diverges from the original intentions of its creators; however, without these elements, the approach has diminished power and longevity. For example, agile is claimed to be adopted, but only the pair programming and story card techniques are employed.

Programming (XP) approach on all projects it undertakes. This includes techniques such as pair programming, story cards, shared code ownership, and self-determining teams. Agile has been in place for more than a decade and is heavily embedded in the day-to-day operations of the company.

Data analysis

Structured coding and analysis of the interview data were conducted in order to identify evidence of management fashion characteristics. Each interview transcript was reviewed by the first author to identify the existence of the five fashion indicators noted in Table 1, as they apply to agile development (see Table 3). Where an interviewee comment mapped to one or more of the indicators, it was highlighted and coded using NVivo. A total of 235 segments of text were identified and coded against the management fashion indicators. In order to verify the validity of the coding, the second author reviewed the complete coding results; all disagreements were discussed and resolved.

Following the coding, we first examined the individual case studies for evidence of management fashion indicators that were particularly prevalent or notably absent, based on the identified interview segments. We re-reviewed the data within each indicator category and began to generate a preliminary 'profile' of each case's configuration of management fashion indicators (e.g. numerous examples of innovation devolution, but few examples related to the norm of rationality). We considered not only the total quantity of interview segments coded to a particular indicator, but also the consistency among interviewees.

Next, we compared the results across cases by looking for patterns within the preliminary profiles that we had created for each individual case. This approach helped us

to refine our initial assessment of the data by enabling a comparison of each management fashion indicator across the cases. During this review, we drew on past approaches by Bergeron *et al* (2004) and Reich & Benbasat (2000) to assign ratings on the extent that each case contained evidence of the fashion indicators. Where the data strongly supported the presence of an indicator (e.g. frequent interviewee references to an indicator, corroboration of perspectives related to an indicator across interviewees, detailed and extensive examples provided in relation to an indicator), we rated the indicator as 'High'; where some data supported the existence of an indicator (e.g. reference to an indicator in only some interviews, minor inconsistencies of indicator perspectives across interviewees, general indicator descriptions and examples provided), we rated the indicator as 'Moderate'; where little or no data supported the existence of an indicator (e.g. few interviewee references to an indicator, significant inconsistencies in indicator perspectives across interviewees, very limited indicator descriptions and examples provided), we rated the indicator as 'Low'. It is important to note that the indicator ratings should not be interpreted as a judgment on the value, effectiveness, or importance of an innovation's adoption; for example, a 'High' rating should not be equated with either a 'good' or 'bad' assessment, but rather that the agile adoption exhibited evidence that was consistent with the presence of one management fashion indicator. We present the results in the next section, followed by a discussion that can aid in distinguishing between the identified patterns.

Results

A summary of the ratings for each management fashion indicator by case study is listed in Table 4. Two cases contained moderate or high ratings for four of the

indicators, two cases contained moderate or high ratings for three of the indicators, and one case contained high ratings for two of the indicators. These results support the argument that agile development is broadly consistent with at least some elements of a management fashion for each of the five cases. However, because of the variance in the results, we argue that simply labeling agile as a transitory fashion fails to sufficiently differentiate how the organizations we studied adopted and used the approach.

In examining the Table 4 data, we first considered the rankings within each case that corresponded with relatively extreme viewpoints (i.e. high or low ratings), which could distinguish the company's adoption and use of agile from the other cases. We then compared these extreme rankings across cases to determine if any simple patterns existed, such as all high or low ratings. As no clear pattern emerged from this approach, we then considered more complex patterns, such as a consistent mix of high and low ratings across the fashion indicators. We also began to consider the role of moderate rankings in discerning patterns among the data. Using an iterative approach, we began to construct possible groupings of cases on the basis of patterns across the five indicators. Where a possible grouping was identified (e.g. both HealthOrg and InsureCorp exhibited a High rating for Norms of Progress), we re-examined the qualitative data to verify if evidence of cross-case similarities existed. Where Moderate ratings were assigned, we considered whether the case data were consistent with the data from other cases that received a high/low rating or was distinct and should be grouped separately.

Table 4 Case study results by management fashion indicator

	<i>Norm of progress</i>	<i>Norm of rationality</i>	<i>Socio-psych. forces</i>	<i>Techno-economic forces</i>	<i>Innovation devolution</i>
<i>LargeMan</i>	Moderate	High	Moderate	High	Moderate
<i>InfoOrg</i>	Low	Moderate	Moderate	Moderate	Moderate
<i>HealthOrg</i>	High	Low	Low	Low	High
<i>InsureCorp</i>	High	Low	Low	Moderate	High
<i>CodeCorp</i>	Low	High	High	Moderate	Low

The authors discussed the groupings and made iterative refinements to the categories based on the review of the underlying data until each case was deemed to be grouped with other cases that exhibited similarities across each of the five fashion indicators. Figure 1 illustrates the approach we adopted.

One organization, CodeCorp, was found to demonstrate either high or moderate evidence of the norms of rationality, socio-psychological, and techno-economic indicators, but virtually no evidence of the other categories. In comparison, two organizations, LargeMan and InfoOrg, exhibited somewhat similar moderate/high categories as CodeCorp, but showed more evidence of the norms of progress and innovation devolution characteristics. Finally, HealthOrg and InsureCorp were essentially the opposite to CodeCorp, with high evidence of norms of progress and innovation devolution, but relatively little evidence from the other categories.

Based on common patterns of the management fashion indicators that emerged across the cases, we propose three distinct categories of agile development adoption and use (see Table 5). We refer to these categories as the Crusader, the Tailor, and the Dabbler, each of which consists of a distinct configuration of management fashion indicator characteristics. Each of the five case studies is proposed as an example of the category that it is most consistent with. Similar to the indicator ratings above, we continue to impose no value judgments on the categories, but present them as an identified pattern of management fashion indicators within the empirical data. Following this, we discuss the link between the fashion indicators, mindfulness/mindlessness, and transitory/pervasive innovations.

The first category, the Crusader, is characterized by high norms of rationality, socio-psychological forces, and techno-economic forces, but low norms of progress and innovation devolution. We view Crusaders as employing a highly prescriptive adoption of agile techniques, alongside an avoidance of traditional approaches entirely. CodeCorp is viewed as a Crusader in its 'by-the-book' adoption of XP and exclusive use of agile techniques. In discussions with staff at CodeCorp, there was broad agreement that agile was a natural fit to the organization's culture and structure (i.e. socio-psychological and techno-economic forces).

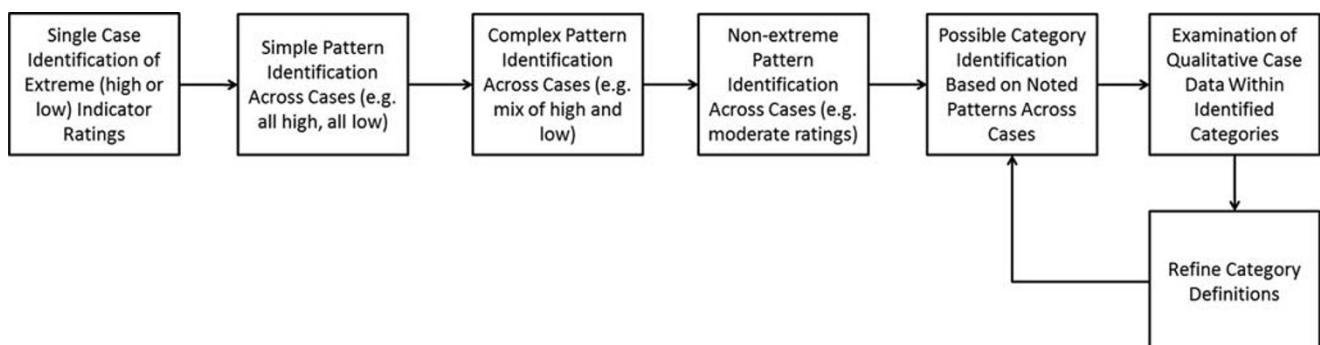


Figure 1 Category identification approach.

Table 5 Categories of agile development adoption and use

Agile adoption & use category	Agile adoption & use characteristics	Management fashion indicators	Case study example(s)
Crusader	Prescriptive adoption of the core agile techniques with minimal adaptation; exclusive use of agile with no other approaches (e.g. waterfall) used; consistent management and staff buy-in to the benefits; strong perceptions of agile fit with the organizational structure and culture; long-term view of agile as an exclusive development approach.	Low norm of progress, high norm of rationality, high socio-psychological forces, high techno-economic forces, low innovation devolution	CodeCorp
Tailor	Adopts most agile techniques, but there is flexibility to make modifications on a project-by-project basis; traditional development approaches are used on some organizational projects or are integrated together with agile; most managers and staff buy in to agile; mid- to long-term view of agile as an adapted or hybrid approach.	Moderate norm of progress, moderate norm of rationality, moderate socio-psychological forces, moderate techno-economic forces, moderate innovation devolution	InfoOrg, LargeMan
Dabbler	Adopts only a few agile techniques, while maintaining a core base of plan-driven development techniques; some employee and management support, but also some open opposition; uncertainty about the organizational and cultural fit of agile; short-term view of agile.	High norm of progress, low norm of rationality, low socio-psychological forces, low techno-economic forces, high innovation devolution	InsureCorp, HealthOrg

There was also a strong sense that agile was a better, more rational approach compared to traditional methods (i.e. norm of rationality). The following interviewee quotes highlights this view:

It's just a much more productive and effective environment. So basically, software development is very much an art, and the ability to leave the artists alone to work on their art, while at the same time being able to facilitate them to do what they do and protect them from all the other outside forces. I think is a real motivator for them. So, it allows the developers to do what they do best and when they're in that mode, they're most productive. (Project Manager)

I think that a major point is that the feedback cycle is a lot shorter. Let's say you are at a stand up and you are discussing what you have been doing over the last two or three days. If you are not going any further then you catch it at that point and you don't catch it two or three weeks down the line. (Developer)

There was little evidence at CodeCorp that suggested either management or staff did not have full confidence in the agile approach. In fact, interviewees highlighted improved morale, fulfillment, and creativity within the development staff. In contrast to past management fashion theory views that frame the norm of rationality indicator as relatively superficial, CodeCorp staff appeared to be more consistent and persuasive in their argument that agile actually is a more effective means of developing software and contributes to the best interests of the organization. This uniformity of employee opinion that agile was clearly a better development approach than the alternatives was a key distinguishing

factor of the Crusader category. The following quote explains this perspective:

The work that is created is a lot better quality and the overall process ends up being faster, because you're kept on task because you're working with other people and you're also [employing a] two heads are better than one kind of idea. So you end up creating a product that's better and you end up getting it done faster in the end. (Developer)

Crusaders are also characterized by an absence of adoption for adoption's sake (i.e. norms of progress) or a desire to concurrently adopt other development approaches (i.e. innovation devolution), which was the case at CodeCorp. Employees at the organization tended to have deep technical experience and extensive familiarity with agile techniques, which could account for their strong opinions. The following quote highlights the Crusader view of adopting a 'pure' form of agile:

Agile has interlocking practices or principles [and] you have to do all of them, right? So, you have to do pair programming, we have to do the testing, we have to do all those pieces in concert. If you try and skip out on one of them, then what ends up happening is [that] you've taken out something that's important ... if you start throwing those things out, then you're in trouble. (Developer)

The Tailor category is characterized by a moderate degree of all five management fashion indicators. Although Tailors adopt many agile techniques, there is an implicit flexibility to adapt some aspects of the approach on a project-by-project basis alongside practices from other development approaches. At LargeMan and InfoOrg, agile was widely adopted, but was used in conjunction

with some elements of a waterfall approach. Our data indicate that management and staff generally agreed with the philosophy and benefits of agile (i.e. norm of rationality), but maintained that the characteristics of the development team, company culture and organizational structure (i.e. socio-psychological forces, techno-economic forces) required some refinements and adjustments to the agile practices. This tended to result in a piecemeal adoption of agile, where select practices became optional for developers. For example, some developers at LargeMan resisted participating in the agile practice of pair programming, where two staff members work together in completing a development task. Management typically permitted pair programming to be conducted either on a voluntary basis or required only a minimum level of adherence to the practice, as the following interviewee quote highlights:

So we have put a challenge out to the whole team that everyone needs to at least do one pair programming session through this next sprint and it is a short sprint, only 2 weeks. So everyone agreed; we will see how it goes. (Development Manager, LargeMan)

A similar circumstance occurred at InfoOrg, where agile project teams utilized weekly meetings, rather than daily stand-up meetings. This decision was made on the basis that staff didn't perceive a sufficient benefit to result from the act of meeting on a daily basis. Although staff were aware that this deviated from agile guidelines, they determined that it was a better fit for their project circumstances. The following quote highlights this approach:

Not daily stand ups ... we usually do more of a weekly thing. Yeah, and that tends to work better for us. I think that we tried doing daily stand up meetings and there just wasn't much point to it. (Developer, InfoOrg)

In addition to the selected adoption of agile practices, perceived misalignments with socio-psychological forces and techno-economic forces also drove the continued use of plan-driven, waterfall approach elements. These factors were most commonly associated with a reluctance to fully depart from a traditional development approach. Although employees appeared to genuinely value the social and technical benefits that agile introduced, they also continued to value elements of their previous development approach. The following quotes provide examples from the LargeMan and InfoOrg case studies:

I guess the major deviation that I am seeing right now is that within a sprint we are not saying that the definition of 'done' is production-ready code. It is code that is completed, but we still run a QA cycle in tandem with every development cycle ... I see that as a deviation from pure Scrum. (Development Lead, LargeMan)

We had a project that we ran sort of [a] half-and-half approach. The base of the project was based on waterfall, because there were some grey areas in the technology program. The technology team basically was gathering every morning ... in the Scrum discussing what they wanted to do that day and what are their problems ...

so it was sort of a mixture of both [approaches]. (Developer, InfoOrg)

The third category that emerged from our analysis is the Dabbler, which is characterized by low norms of rationality, socio-psychological forces, and techno-economic forces, but high norms of progress and innovation devolution. Dabblers typically adopt and use agile in a largely ceremonial way that employs a few agile techniques, but remains primarily rooted in a plan-driven development approach. Both InsureCorp and HealthOrg are seen to belong to this category. Although some employee and management support for agile existed in both organizations, the push to adopt the approach appears to be driven by management's desire to be perceived as adopting an innovative tactic to address challenges with the current development approach (i.e. norms of progress). Few justifications were made by interviewees about how agile was viewed as being a good fit for the organization or staff; rather, agile was most commonly discussed in general terms as being new and better, such as with the following quote:

Everyone is willing and interested in how can we do things better because we are finding that we need to get better. It is an important capability for our organization. There is a lot of change out there. (Development Manager, InsureCorp)

There was also more open opposition to the agile approach than there was in the Crusader or Tailor categories, which is indicative of misalignment between agile and the corresponding socio-psychological/techno-economic forces. At both InsureCorp and HealthOrg, there was a degree of uncertainty about what agile actually represented and a lack of clarity about whether the approach fit with the culture and structure of the organization. In contrast to employees within a Crusader company, Dabbler employees tended to have less development experience and acted as technology generalists rather than experts in a particular area. This unfamiliarity with agile and lack of opinion on how it may help is consistent with a much shorter-term, transitional view of agile, as the following quotes highlight:

Given how small we are, I would not be surprised if, as we learned a little bit more about agile, that we either make a decision to stick with it or drop it. (IT Director, HealthOrg)

I think that agile is useful in some situations and not so useful in others and then there are other sorts of hybrid approaches that we should consider as well. But I don't think that there is any universal answer and one-size-fits-all sort of SDLC methodology. (Development Manager, InsureCorp)

Dabblers were distinct from Tailors in their degree of innovation devolution. Rather than assembling a hybrid development approach that merged the use of agile with some plan-driven practices, Dabblers had much more difficulty leaving their plan-driven traditions behind. This was often the case because many agile techniques

fundamentally diverged from the tenets of traditional development approaches (e.g. requirements, documentation, etc.) and following the agile guidance was viewed by employees as being inconsistent with commonly accepted conventions. Interviewees suggested that developers resisted agile techniques they viewed as being at odds with established practices:

Talking to my PM who is actually leading this process, the staff are holding on to the fact that they need requirements and they still are falling back on the fact that they need lots of documentation. (Project Management Director, InsureCorp)

So in order for us to implement something, we still have to go through our management process of getting it approved and a go-live and a sign off by the directors and we have to prove that we have tested and that it has been accepted. (Development Manager, HealthOrg)

This section has presented three proposed categories of agile adoption and use that emerged from our analysis of management fashion indicators. Based on our collected data, we have assigned each of the five participating companies to the category its operations are most consistent with. Next, we discuss these findings in the context of the agile, management fashion, and mindfulness literature.

Discussion

As noted above, the organizations grouped as Crusaders, Tailors, or Dabblers were influenced differently by the management fashion indicators. Whereas Crusaders and Tailors were influenced mostly by norms of rationality, socio-psychological forces, and techno-economic forces, Dabblers were influenced most strongly by characteristics of fashions related to the norm of progress and innovation devolution. Based on these patterns, this section discusses how our data support insights into how management fashion concepts aid in better understanding the mindful (or mindless) and more permanent (or transitory) adoption of agile.

Drawing on the mindfulness and mindlessness characteristics identified in past research related to general innovations (Swanson & Ramiller, 2004; Butler & Gray, 2006), as well as agile (Matook & Kautz, 2008; McAvoy *et al*, 2013), we re-reviewed the three categories and the associated empirical data. In examining the patterns within these categories, distinct perspectives were uncovered relating to mindful vs mindless agile adoption: both Crusaders and Tailors were found to share characteristics that exhibit mindfulness (albeit in different ways), while Dabblers exhibited characteristics most commonly (though not exclusively) associated with mindlessness. However, we noted that exceptions exist that could lead Crusaders to exhibit mindlessness and Dabblers to be mindful. For example, CodeCorp (a Crusader) was highly reluctant to simplify practices and paid close attention to methodological details, two characteristics commonly attributed to mindfulness (Swanson & Ramiller, 2004). They were adamant in their refusal to adapt or alter their pure agile

approach, which they interpreted as an 'all or nothing' proposition; this is consistent with minimal innovation devolution from a fashion perspective. CodeCorp also exhibited the mindfulness trait of deference to expertise and the avoidance of formal structure (Swanson & Ramiller, 2004) through their flat organizational structure and shared roles on project teams, which corresponds to elements of socio-emotional and techno-economic indicators. Although the approach of Crusaders, then, can be viewed as being largely consistent with the concept of mindfulness, we can also identify a potential weakness for this type of fashion adopter. Where adopters avoid making changes in order to maintain a fashionable innovation's complete value, it would be seen as being consistent with past views of mindfulness that promote a reluctance to simplify and a commitment to resilience (Swanson & Ramiller, 2004; McAvoy *et al*, 2013). However, in cases where the innovation is (or becomes) a poor fit with the organization's characteristics (i.e. socio-emotional and techno-economic forces), Crusaders may stubbornly avoid adaptation (i.e. innovation devolution), even in circumstances where the technique is failing. This inattention to the specifics suggests a mindless rather than a mindful adoption.

Tailors such as InfoOrg and LargeMan exhibited the mindfulness characteristics of resiliency and alertness to failure (Swanson & Ramiller, 2004). Although they were willing to take risks by moving away from plan-driven approaches in favor of an agile approach, Tailors were continually refining and evaluating the effectiveness of agile. Although agile did not work perfectly every time, customizations were made to improve and adapt the approach going forward. These customizations can be viewed as a moderate degree of innovation devolution, as a completely pure agile approach was not being employed. However, these modifications were undertaken on the basis that the adapted version of agile was perceived as a better fit for the organization (i.e. techno-economic forces) and its staff (i.e. socio-psychological forces). Indeed, there was little evidence that agile was in jeopardy of being abandoned at Tailor organizations in the short term. We found Tailors to exhibit the most mindfulness characteristics, alongside little evidence of mindlessness. In contrast to Crusaders who maintained a 'pure' agile approach, Tailors appeared to demonstrate a greater awareness of their own organizational circumstances, making the adjustments required by the methodology to fit their particular situation. A potential drawback of Tailors from a mindfulness perspective could result from a circumstance that awareness of organizational circumstances results in adjustments of an innovation that extend beyond tailoring to deteriorate into innovation devolution. For example, where the extent of adjustments made to a technique substantively changes its spirit from the original intent, it may no longer contain the same degree of value (Zbaracki, 1998; Benders & van Veen, 2001), which may limit the organizational benefits received.

In contrast, Dabblers exhibited characteristics more consistent with mindlessness, often seeming to follow the promises of success stories that praise agile as the latest and

greatest trend, which is consistent with the norm of progress fashion indicator. They were generally anxious to simplify the agile approach wherever possible (e.g. weekly, rather than daily meetings), paid minimal attention to the nuances of the approach (e.g. dismissal of pair programming as a valuable practice), and appeared to have little invested in the long term success of agile (e.g. minimal staff training), which is consistent with a high degree of innovation devolution. Despite the limitations that this type of adoption and use has on the ability to deliver strong organizational benefits, Dabbling may still represent a viable choice for organizations since they can accrue the political and reputational benefits that prior research suggests can come from fashions (Wang, 2010). Even though Dabblers may not generate the operational gains anticipated by Crusaders or Tailors, the adoption of agile could still signal to stakeholders (e.g. executives, customers) that the systems development process is embracing modern advances. However, in contrast to Crusaders and Tailors that exhibited longer-term plans for agile, Dabblers appear to be focused on short-term gains, without investing the time and money to fully appropriate the innovation. Although some organizations are likely to be mindlessly jumping from one fashion to the next without considering the organizational drivers and drawbacks, it is possible that a mindful Dabbler may gain short-term benefits at a minimal cost. Future research could examine if, at the point that agile begins to wane in terms of prestige and perceived value, a mindful Dabbler could adopt a new innovation that renewed the political and reputational advantages that fashions provide. One possible explanation for the Dabbler category is that some organizations adopting agile may need to undergo an initial period of learning while they transition away from traditional development techniques. Where this transition is able to occur, agile development may become further embedded and the organization could transition into the Crusader or Tailor category. In other cases, the approach may be abandoned in favor of a newly emerging fashion. It is unclear which path our two Dabblers may take; agile is a relatively new technique at HealthOrg, while InsureCorp has already adopted, abandoned, and re-adopted the approach. Although the cross-sectional nature of our data limits the extent that we can identify changing adoption and use patterns over time, future studies that employ a longitudinal approach could uncover important insights that build on our initial observations.

The Dabbler category is most consistent with the latency-popularity-decline stage model that management fashions are largely seen to follow (Abrahamson & Fairchild, 1999; Carson *et al*, 1999). When it appears that the anticipated benefits promised by agile are waning, the necessity for alternatives or counter-fashions will increase (Kieser, 1997). Because of the conceptual ambiguity and flexibility inherent in agile, particularly for Tailors and Dabblers, our results suggest that it may provide opportunities for organizations to adapt the innovation for their particular needs, which could diminish the demand for

counter-fashions. However, it also allows companies the flexibility to only ceremonially adopt agile, without adopting its core philosophy or many of its techniques. If, for example, more organizations adopt agile as Dabblers and the subsequent performance results are below expectations, it may provide an opportunity for the gurus, consultants, and academics to seek out and propose a new, alternative development approach. This view is compatible with Benders and van Veen's (2001) suggestion that fashions may also decline in popularity as a result of such extensive adoption that fashion followers can no longer distinguish themselves from others. If the rise of agile continues to grow, it may be viewed as less distinctive and politically valuable for both mindless and mindful Dabblers, who could abandon it for more novel fashions.

Following the view that fashions represent transitory beliefs that are iteratively adopted based on the dissemination of management knowledge (Abrahamson, 1991; Abrahamson, 1996), the characteristics of the Dabbler type appears to fit this view most effectively. Both InsureCorp and HealthOrg had distinctly short-term views of agile and exhibited a high degree of innovation devolution and attraction to agile as an embodiment of 'progress'. In contrast, the Crusader and Tailor appeared less consistent with the perception of agile as a passing fashion, where the adoption of agile was made on the basis of more mindful, reasoned arguments that considered longer-term use of the development approach. Because of this longer-term view of agile at CodeCorp, LargeMan, and InfoOrg, it would suggest that agile is increasingly viewed as an approach that has a degree of pervasiveness. Carson *et al* (1999) suggest that the progression of a fashion into established collective wisdom occurs when a technique's benefits outweigh the costs in the short and long term, implementation is not more difficult than expected, the technique improves the organization's status, and there is institutional support for the technique. Based on the empirical evidence, it appears that these requirements are largely consistent with Crusaders and Tailors, though not with Dabblers. Rather, our results challenge the argument made by Carson *et al* (1999) by suggesting that in addition to the macro-level characteristics of a fashion, a management technique's ability to become pervasive within an organization is also shaped by the extent that fashion indicators have influenced the adoption at the firm level.

Although a growing body of research supports the view that the flexibility of hybrid agile-waterfall approaches employed by Tailors are a viable alternative to an unmodified, agile-only approach (Vinekar *et al*, 2006; Port & Bui, 2009; West & Grant, 2010), it highlights the relatively extreme views of Crusaders. In contrast to the anticipated depletion of performance that the management fashion literature would suggest when moving away from a 'pure' form of agile (i.e. a devolution of the innovation), systems development research suggests that the Tailor may be no worse off than the Crusader and in some cases it may even be preferable (Fitzgerald *et al*, 2006; Cao *et al*, 2009). In relating this observation to the role of industry, our

lone Crusader (CodeCorp), specializes in software development, while the Tailors (InfoOrg, LargeMan) and Dabblers (InsureCorp, HealthOrg) represent the media, manufacturing, insurance, and healthcare industries. In addition to suggesting that Crusaders may be less common compared to Tailors and Dabblers, it also highlights the characteristics that may distinguish Crusaders, such as in-house development expertise, IT thought leadership, and a desire to adopt innovative development practices. Although these characteristics could well exist at companies within a range of industries, it may suggest that Crusaders are more prominent in technology-driven, software-centric industries.

In the context of the three categories outlined above, it is possible that mindfulness could be enhanced in circumstances where the choice of agile adoption most closely fit the organizational context. We see this as most applicable to Crusaders, who fully adopt a pure agile approach. Future studies could evaluate if mindful adoptions are consistent with an organization pursuing a Crusader approach when their circumstances are consistent with Boehm & Turner's (2003) ideal agile factors (e.g. small teams, turbulent environment). Alternatively, a Crusader's adoption of agile in circumstances consistent with plan-driven factors (e.g. large teams, stable environment) may be indicative of a mindless approach, due to the inflexibility that Crusaders employ in implementing the development approach.

By considering the insights noted above that are drawn from our analysis, a series of more general, theory-focused insights can be extracted. First, our results suggest that a variety of paths can lead to mindful and mindless adoption of innovations. This indicates that the transitory and partial *vs* more permanent and holistic adoption of a new management practice does not equate directly to the mindless *vs* mindful distinction. As a result, managers can benefit by becoming increasingly aware of the pitfalls, as well as the opportunities, associated with how they adopt and use new innovations. This awareness can help to guide decisions related to the time, money, and training that should be allocated towards a new trend *vs* an existing approach. Second, the nature of an organization's adoption and use of a management technique is directly linked to the fundamental decisions that will guide the innovation towards increased benefits or declining value. Where an organization falls within the Dabbler category, managers should seek to avoid blindly following each new trend, but instead seek out new innovations that can iteratively enhance the political and reputational benefits of the latest fashion. In contrast, Crusaders and Tailors should aim to exhibit mindful characteristics by dedicating resources towards the refinement and institutionalization of management techniques for the mid to long term. However, Crusaders should be particularly careful to avoid mindlessly pursuing innovations where organizational circumstances may indicate that a hybrid, Tailor-style approach may be more suitable for its circumstances.

Conclusions

The objective of this study was to examine how management fashion concepts aid in better understanding how organizations mindfully (or mindlessly) adopt an agile development approach. Although mindfulness in adopting agile has been considered in prior research, we sought to understand the practical links to management fashions at the firm level. Drawing on 42 interviews across five case studies, we examined the extent that five management fashion indicators were present: norms of progress, norms of rationality, socio-psychological forces, techno-economic forces, and innovation devolution. Based on patterns emerging from the management fashion indicators in each case study, we identified three categories of agile adoption and use: Crusaders, who exclusively employ agile in a pure form; Tailors, who customize agile to fit their specific circumstances, while concurrently employing traditional techniques; and Dabblers who employ a few ceremonial agile techniques on top of a fundamentally traditional development approach. We consider those organizations falling into the Dabbler category as being more likely to adopt agile as a short-term, transitory approach before moving on to the next development trend that emerges. In contrast, our findings suggest that organizations within the Crusader and Tailor categories are more likely to maintain agile development beyond the short term, with the development approach becoming an increasingly pervasive organizational practice.

Limitations and future research

As with any study, our work has a number of limitations and opportunities for future research. First, the cross-sectional data used in our study limits the breadth of understanding we have about how management fashions change over time. We recognize the benefits that could be gained by conducting longitudinal research to examine the extent that an organization may evolve in its adoption and use of a management technique, perhaps even transitioning from one category to another (e.g. Dabbler to Tailor). This line of inquiry could also be useful in companies that have adopted, abandoned, and then re-adopted an agile approach, like at InsureCorp. Our research is also limited by the five management fashion indicators that formed the basis for our analysis, as there may be additional indicators or unique patterns that could further enhance our understanding of the mindful adoption and use of innovations. For example, although we noted the three distinct patterns of management fashion indicators listed in Table 5, it is possible that additional configurations of these indicators could exist in other organizations. To this end, although we argue that concepts from management fashion theory contribute to developing our understanding of how agile is adopted and used, it is one of many available perspectives. Future research could consider alternative theoretical approaches that complement the insights provided by fashions, including self-efficacy theory, structuration theory, and resource dependency theory. Finally, although we

attempted to select a range of cases in order to get a varied view of agile adoption and use, extending the sample of participants would be beneficial. This could include a wider range of agile types (e.g. three of five cases used Scrum, while only one used XP), as well as further investigation on the possible links between industry and Crusader/Tailor/Dabbler categorization (e.g. frequency of Crusaders in the software industry vs the manufacturing industry). This could further refine our model and identify new patterns that exist in relation to our agile adoption and use categorizations.

About the Authors

W. Alec Cram is an assistant professor of Information and Process Management at Bentley University. His research interests focus on how information systems governance and control initiatives can contribute to improving the performance of organizational processes, such as systems development and information security.

The insights from this study contribute to the growing research on agile development by applying a management fashion theory lens that can aid in better understanding how agile is adopted and used. Our findings suggest that although some agile adoptions exhibit characteristics of mindlessness and are consistent with passing fashions, others exhibit evidence of a mindful, pervasive role for agile development in organizations. Our proposed categories highlight the characteristics that distinguish between agile adoptions, while shedding light on the paths that can lead to mindful and mindless innovations.

Sue Newell is a professor in the Department of Business and Management at the University of Sussex. Her research focuses on innovation, specifically on understanding how knowledge is transferred and innovation fostered within and across organizations.

References

- ABRAHAMSON E (1991) Managerial fads and fashions: the diffusion and rejection of innovations. *Academy of Management Review* **16**(3), 586–612.
- ABRAHAMSON E (1996) Management fashion. *Academy of Management Review* **21**(1), 254–285.
- ABRAHAMSON E and FAIRCHILD G (1999) Management fashion: lifecycles, triggers, and collective learning processes. *Administrative Science Quarterly* **44**(4), 708–740.
- ABRAHAMSSON P, CONBOY K and WANG X (2009) 'Lots done, more to do': the current state of agile systems development research. *European Journal of Information Systems* **18**(4), 281–284.
- BALIJEPPALLY V, MAHAPATRA R, NERUR S and PRICE KH (2009) Are two heads better than one for software development? The productivity paradox of pair programming. *MIS Quarterly* **33**(1), 91–118.
- BASKERVILLE RL and MYERS MD (2009) Fashion waves in information systems research and practice. *MIS Quarterly* **33**(4), 647–662.
- BECK K *et al* (2001) 'Manifesto for agile software development.' Retrieved August 2014, from [WWW document] <http://agilemanifesto.org>.
- BENDERS J and VAN VEEN K (2001) What's in a fashion? Interpretive viability and management fashions. *Organization* **8**(1), 33–53.
- BERGERON F, RAYMOND L and RIVARD S (2004) Ideal patterns of strategic alignment and business performance. *Information & Management* **41**(8), 1003–1020.
- BOEHM B and TURNER R (2003) Using risk to balance agile and plan-driven methods. *Computer* **36**(6), 57–66.
- BUTLER BS and GRAY PH (2006) Reliability, mindfulness, and information systems. *MIS Quarterly* **30**(2), 211–224.
- CAO L, MOHAN K, XU P and RAMESH B (2009) A framework for adapting agile development methodologies. *European Journal of Information Systems* **18**(4), 332–343.
- CARSON PP, LANIER PA, CARSON KD and BIRKENMEIER BJ (1999) A historical perspective on fad adoption and abandonment. *Journal of Management History* **5**(6), 320–333.
- CHOW T and CAO D-B (2007) A survey study of critical success factors in agile software projects. *The Journal of Systems and Software* **81**(6), 961–971.
- CLARK T (2004) The fashion of management fashion: a surge too far? *Organization* **11**(2), 297–306.
- CONBOY K (2009) Agility from first principles: reconstructing the concept of agility in information systems development. *Information Systems Research* **20**(3), 329–354.
- CONBOY K and FITZGERALD B (2010) Method and developer characteristics for effective agile method tailoring: a study of xp expert opinion. *ACM Transactions on Software Engineering and Methodology* **20**(1), 1–30.
- DE CESARE S, LYCETT M, MACREDIE RD, PATEL C and PAUL R (2010) Examining perceptions of agility in software development practice. *Communications of the ACM* **53**(6), 126–130.
- DINGSOYR T, NERUR S, BALIJEPPALLY V and MOE NB (2012) A decade of agile methodologies: towards explaining agile software development. *Journal of Systems and Software* **85**(6), 1213–1221.
- DRURY M, CONBOY K and POWER K (2012) Obstacles to decision making in agile software development teams. *The Journal of Systems and Software* **85**(6), 1239–1254.
- DUBE L and PARE G (2003) Rigor in information systems positivist case research: current practices, trends, and recommendations. *MIS Quarterly* **27**(4), 597–635.
- DYBA T and DINGSOYR T (2008) Empirical studies of agile software development: a systematic review. *Information and Software Technology* **50**(9–10), 833–859.
- FIOL CM and O'CONNOR EJ (2003) Waking up! mindfulness in the face of bandwagons. *The Academy of Management Review* **28**(1), 54–70.
- FITZGERALD B, HARTNETT G and CONBOY K (2006) Customising agile methods to software practices at intel Shannon. *European Journal of Information Systems* **15**(2), 200–213.
- HIRSCHHEIM R, MURUNGI DM and PENA S (2012) Witty invention or dubious fad? Using argument mapping to examine the contours of management fashion. *Information and Organization* **22**(1), 60–84.
- HOLMSTROM H, FITZGERALD B, AGERFALK PJ and CONCHUIR EO (2006) Agile practices reduce distance in global software development. *Information Systems Management* **23**(3), 7–18.
- JANES A and SUCCI G (2012) The dark side of agile software development. *Proceedings of the ACM International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software*, 19–26 October, Association for Computing Machinery, Tucson, Arizona.
- KIESER A (1997) Rhetoric and myth in management fashion. *Organization* **4**(1), 49–74.
- LAANTI M, SALO O and ABRAHAMSSON P (2011) Agile methods rapidly replacing traditional methods at Nokia: a survey of opinions on agile transformation. *Information and Software Technology* **53**(3), 276–290.
- LANGER EJ (1989a) *Mindfulness*. Addison-Wesley, Reading, MA.
- LANGER EJ (1989b) Minding matters: the consequences of mindlessness-mindfulness. *Advances in Experimental Social Psychology* **22**, 137–173.
- LANGER EJ (1997) *The Power of Mindful Learning*. Addison-Wesley, Reading, MA.
- LANGER EJ, BLANK A and CHANOWITZ B (1978) The mindlessness of ostensibly thoughtful action: the role of 'placebic' information in interpersonal interaction. *Journal of Personality and Social Psychology* **36**(6), 635–642.

- LAWLER EEI and MOHRMAN SA (1985) Quality circles after the fad. *Harvard Business Review* **63**(1), 64–71.
- LEVINTHAL D and RERUP C (2006) Crossing an apparent chasm: bridging mindful and less-mindful perspectives on organizational learning. *Organization Science* **17**(4), 502–513.
- MANGALARAJ G, MAHAPATRA R and NERUR S (2009) Acceptance of software process innovations – the case of extreme programming. *European Journal of Information Systems* **18**(4), 344–354.
- MARUPING LM, ZHANG X and VENKATESH V (2009) Role of collective ownership and coding standards in coordinating expertise in software project teams. *European Journal of Information Systems* **18**(4), 355–371.
- MATOOK S and KAUTZ K (2008) Mindfulness and agile software development. In: *19th Australasian Conference on Information Systems*. Christchurch.
- MCAVOY J, NAGLE T and SAMMON D (2013) Using mindfulness to examine Isd agility. *Information Systems Journal* **23**(2), 155–172.
- MEYER JW and ROWAN B (1977) Institutionalized organizations: formal structure as myth and ceremony. *American Journal of Sociology* **83**(2), 340–363.
- MOCZAR L (2013) Why agile isn't working: bringing common sense to agile principles. CIO, June 4, [WWW document] <http://www.cio.com/article/2385322>.
- MU E and BUTLER BS (2009) The assessment of organizational mindfulness processes for the effective assimilation of it innovations. *Journal of Decision Systems* **18**(1), 27–51.
- NERUR S and BALIJEPPALLY V (2007) Theoretical reflections on agile development methodologies. *Communications of the ACM* **50**(3), 79–83.
- NERUR S, MAHAPATRA R and MANGALARAJ G (2005) Challenges of migrating to agile methodologies. *Communications of the ACM* **48**(5), 73–78.
- NEWELL S, SWAN J and KAUTZ K (2001) The role of funding bodies in the creation and diffusion of management fads and fashions. *Organization* **8**(1), 97–120.
- NEWELL S, SWAN J and ROBERTSON M (1998) A cross-national comparison of the adoption of business process reengineering: fashion-setting networks? *Journal of Strategic Information Systems* **7**(4), 299–317.
- PORT D and BUI T (2009) Simulating mixed agile and plan-based requirements prioritization strategies: proof-of-concept and practical implications. *European Journal of Information Systems* **18**(4), 317–331.
- RAMILLER NC, SWANSON EB and WANG P (2008) Research directions in information systems: toward an institutional ecology. *Journal for the Association for Information Systems* **9**(1), 1–22.
- REICH BH and BENBASAT I (2000) Factors that influence the social dimension of alignment between business and information technology objectives. *MIS Quarterly* **24**(1), 81–113.
- ROGERS E (1995) *Diffusion of Innovations*. Free Press, New York.
- SENAPATHI M and SRINIVASAN A (2012) Understanding post-adoptive agile usage: an exploratory cross-case analysis. *The Journal of Systems and Software* **85**(6), 1255–1268.
- SHARP H, BIDDLE R, GRAY P, MILLER L and PATTON J (2006) Agile development: opportunity or fad? In *CHI '06 Extended Abstracts on Human Factors in Computing Systems*, 22–27 April Association for Computing Machinery, Montreal, Canada.
- SWANSON EB and RAMILLER NC (1997) The organizing vision in information systems innovation. *Organization Science* **8**(5), 458–474.
- SWANSON EB and RAMILLER NC (2004) Innovating mindfully with information technology. *MIS Quarterly* **28**(4), 553–583.
- VIDGEN R and WANG X (2009) Coevolving systems and the organization of agile software development. *Information Systems Research* **20**(3), 355–376.
- VINEKAR V, SLINKMAN CW and NERUR S (2006) Can agile and traditional systems development approaches coexist? An ambidextrous view. *Information Systems Management* **23**(3), 31–42.
- VOGUS TJ and WELBOURNE TM (2003) Structuring for high reliability: HR practices and mindful processes in reliability-seeking organizations. *Journal of Organizational Behavior* **24**(7), 877–903.
- WANG P (2010) Chasing the hottest it: effects of information technology fashion on organizations. *MIS Quarterly* **34**(1), 63–85.
- WANG X, CONBOY K and PIKKARAINEN M (2012) Assimilation of agile practices in use. *Information Systems Journal* **22**(6), 435–455.
- WEICK KE and SUTCLIFFE KM (2006) Mindfulness and the quality of organizational attention. *Organization Science* **17**(4), 514–524.
- WEST D and GRANT T (2010) Agile development: mainstream adoption has changed agility. *Forrester Research*, pp. 1–20, Cambridge, MA.
- WESTRUP C (2003) *Discourse, Management Fashions, and Erp Systems*. Kluwer Academic Publishers, Norwell, MA.
- YIN RK (2009) *Case Study Research: Design and Methods*. SAGE, Thousand Oaks, CA.
- ZBARACKI MJ (1998) The rhetoric and reality of total quality management. *Administrative Science Quarterly* **43**(3), 602–636.